

CS-1851

Examination - Nov- 2022
B.Tech. V Sem : Computer Science & Engineering
Operating System

Time : 3 Hrs

Max. Marks : 70

Min. Marks : 22

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a, b & c are compulsory while Part d has internal Choice. Assume missing data, if any.

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Word limit NOT to be followed for diagram, numerical, derivation.

- Q.1 (a) What are the objectives of operating system? 02
(b) Timesharing differ from multiprogramming? If so, how? 02
(c) Distinguish between batch systems and time sharing systems. 03
(d) Explain different operating system structures 07

OR

What are system calls? Explain different categories of system calls with example? 07

- Q.2 (a) What are the data fields associated with process control blocks? 02
(b) Distinguish between CPU bounded, I/O bounded processes. 02
(c) Distinguish between i) Process and Program ii) Job scheduling and CPU scheduling 03
(d) Consider the set of 4 processes whose arrival time and burst time are given below- 07

Process No.	Arrival Time	Priority	Burst Time		
			CPU Burst	I/O Burst	CPU Burst
P1	0	2	1	5	3
P2	2	3	3	3	1
P3	3	1	2	3	1

If the CPU scheduling policy is Priority Scheduling, calculate the average waiting time and average turn around time. (Lower number means higher priority) $\frac{25}{3}$

OR

What are co-operating processes? Describe the mechanism of inter process communication using shared memory in a producer-consumer problem. 07

- Q.3 (a) Explain the resource allocation graph
 (b) Describe necessary conditions for a deadlock situation to arise.
 (c) Describe resource allocation graph with a deadlock, with a cycle but no deadlock.
 (d) Given 3 processes A,B and C, three resources x,y and z and following events, a. A requests x ii) A requests y iii) B requests y iv) B requests z v) C requests z vi) C requests x vii) C requests y Assume that requested resources should always be allocated to the request process if it is available. Draw the resource allocation graph for the sequences. And also mention whether it is a deadlock? If it is, how to recover the deadlock.

OR

A system has 4 processes and 5 allocatable resource. The current allocation and maximum needs are as follows-

Allocated					Maximum					
A	1	0	2	1	1	1	1	2	1	3
B	2	0	1	1	0	2	2	2	1	0
C	1	1	0	1	1	2	1	3	1	1
D	1	1	1	1	0	1	1	2	2	0

If Available = [0 0 X 1 1], what is the smallest value of x for which this is a safe state?

- Q.4 (a) Illustrate paging and swapping? 02
 (b) Discuss external fragmentation. What are the causes for external fragmentation? 02
 (c) Explain the difference between Physical and logical address 03
 (d) Consider the reference stream 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6. How many page faults while using FCFS and LRU using 2 frames? 07

OR

What is the various Page Replacement Algorithms used for Page Replacement? Give a Brief description of them. 07

- Q.5 (a) Discuss various layers of a file system. 02
 (b) Illustrate the main purpose of Disk scheduling? 02
 (c) Explain in detail the file system in LINUX system. 03
 (d) Explain the access matrix structure employed in protection domain? 07

OR

Explain the following disk scheduling algorithm with examples. i)SSTF ii) SCAN iii)LOOK Comment on the selection of these scheduling methods 07

Examination - April/May- 2023
B.Tech. IV Sem : CSE-ICB
Operating System

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- Q.1 (a) Explain Multiprocessor Systems & give their advantages. ✓ 02
 (b) Illustrate the Time-sharing Systems. ✓ 02
 (c) Explain difference between Multitasking and Multi Programming. ✓ 03
 (d) Explain different types of system calls with suitable example. ✓ 07

OR

Explain the functionalities of Operating Systems? Describe in detail. ✓ 07

- Q.2 (a) Illustrate various types of schedulers. ✓ 02
 (b) Define Thread and its types. ✓ 02
 (c) Define Process? Explain process State diagram. ✓ 03
 (d) Explain CPU Scheduling Algorithms with examples. ✓ 07

OR

Consider 3 processes P1, P2 and P3, which require 5, 7 and 4 time units and arrive at time 0, 1 and 3. Draw the Gant chart, process completion sequence and average waiting time for. (i) Round robin scheduling with CPU quantum of 2 time units. 07

- Q.3 (a) Illustrate the condition necessary for a deadlock situation to arise. ✓ 02
 (b) State the requirement to be satisfied for a solution of a critical section problem. ✓ 02
 (c) Define race condition. ✓ 03
 (d) Define process synchronization and explain Peterson solution algorithms. ✓ 07

OR

Explain Banker's Algorithm with Example. ✓ 07

- Q.4 (a) Describe Memory Compaction. ✓ 02
 (b) Explain Page Fault. ✓ 02
 (c) Explain segmented memory management. ✓ 03

- (d) Consider the following reference string 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1. 07
Assume there are three frames. Apply LRU replacement algorithm to the reference string above and find out how many page faults are produced.

OR

What is virtual memory? Explain. Suppose we have a demand paged memory. The page table is held in registers. It takes 8ms to service a page fault if an empty page is available or the replaced page is not modified, and 20ms if the replaced page is modified. Memory access time is 100ns. Assume that the page to be replaced is modified 70% of the time. What is the maximum acceptable page fault rate for an effective access time of no more than 200ns? 07

- Q.5 (a) Explain in brief file system mounting operation. 02
(b) What are the three methods for allocating disk space? Explain in Brief. 02
(c) Explain the following (i) file types (ii) file operation (iii) file attributes. 03
(d) Explain the following disk scheduling algorithm with examples. (i) SSTF (ii) SCAN 07

OR

Write short notes on-

- (a) Features of windows- 10.
(b) WINDOWS -10 architecture

Examination - April/May- 2023

B.Tech. IV Sem : Artificial Intelligence & Data Science
Operating System

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- Q.1
- (a) Which type of errors will be handled by the Operating System? 02
 - (b) Write down desirable characteristics of operating system. 02
 - (c) Discuss the pros and cons of buffer cache. 03
 - (d) Discuss different types of operating system with examples of each. 07

OR

What are the functionalities of Operating System? Explain in detail. 07

- Q.2
- (a) What is System Calls? 02
 - (b) Explain the problem solved by Dijkstra Banker's algorithm. 02
 - (c) Explain different criteria for evaluating the CPU scheduling algorithm. 03
 - (d) Find out the average waiting time for the processes arrived in the order x,y,z,w and scheduled under FCFS scheduling with burst time 8,15,10 and 7 respectively. 07

OR

What is the difference between average waiting time of problem solved by FCFS and Non Preemptive SJF for 4 process p1, p2, p3 and p4 and their burst time is 6,8,7,3 respectively? 07

- Q.3
- (a) What is thrashing? 02
 - (b) Differentiate between external and internal fragmentation. 02
 - (c) A cache memory needs an access time of 40 ns and main memory 120 ns, what is the average access time of CPU (assume hit ratio = 80%)? 03
 - (d) In fixed size partitioning if four process comes p1 with size 357, p2 with size 210, p3 with size 468 and p4 with size 491 And vacant memory blocks are 200, 400, 600, 500, 300, 250 Find out the total internal fragmentation in First Fit approach, also which approach satisfy all the processes? 07

OR

Consider the main memory with capacity of 4 page frames. Assume that the pages of a process are referenced in the order as given below:
1,3,4,4,3,2,1,7,5,6,4,2,1,2. Which one is better FIFO or LRU and why?

- Q.4 (a) What is Disk Scheduling? 02
(b) Discuss various file allocation and access methods. 02
(c) Find capacity of disk with 16 platters, 2 surfaces per platter, 1K tracks per surface, 2K sectors per track and 2048 Bytes per sector. 03
(d) Assuming that the disk head is located initially at 30, find the number of disk moves required with FCFS if the disk queue of I/O block requests are 100,40,15,125,73,76. 07

OR

A disk has 300 cylinders (0 to 299). If the initial position of the read-write head is at cylinder 101. Calculate the total head movement by using SSTF disk scheduling algorithm for the read request sequence: 124,46,200,85,240,190,270. 07

- Q.5 (a) What is cryptography? 02
(b) Differentiate between program threats and network threats. 02
(c) What is the difference between worm and virus? 03
(d) How does UNIX provide file protection? Explain. 07

OR

Discuss various security threats in file system of Operating System. 07

Examination – April/May- 2023
B.Tech. IV Sem : Internet of Things
Operating System

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- Q.1 (a) List and state any four services provided by an operating system. 02
- (b) List any four types of operating systems. 02
- (c) Can be ensure same degree of security as in standalone System and timesharing system? 03
- (d) Write the stages of program development and list out the name of system software required in each stage. 07

OR

Compare multitasking, multiprogramming, and multithreading operating system. 07

- Q.2 (a) Define critical section. 02
- (b) List the method of intercrosses communication. 02
- (c) List the features of semaphore. 03
- (d) What is deadlock? Apply the Banker's algorithm on the given data to prevent the deadlock and find 07
- Total no. of available resources in system.
 - Find the Sequence of processes execution if system in safe state.
 - If P1 process resource request is (5, 5, 5) then system will be in safe mode or not.

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	3			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

OR

Describe critical section problem in detail with suitable example. Also explain semaphore and its use. 07

- Q.3 (a) What is MVT & MFT? 02
- (b) Differentiate between segmentation and paging. 02
- (c) Mention the Internal fragmentation in external fragmentation in the context of paging. 03
- (d) Find the total no. of page faults and hit ratio using FIFO, Optimal and LRU page replacement algorithm for the given reference string. (Assuming frame size = 3 & 4 and initially empty). **Reference String = 2,2,0,1,0,7,0,1,5,3,8,5,7,7,4.** 07

OR

At some point in time, the following holes (in the order) are created by a variable partition memory. 20K, 15K, 40K, 60K, 10K, 25K. For some new processes of size 17K, 38K, 23K, 40K, 55K, 20K which holes would be filled using best fit, first fit, and worst fit? How much would be the internal & external fragmentation? 07

- Q.4 (a) Define seek time, latency and access time for a Disk. 02
- (b) List the approaches used in free space management. 02
- (c) Compare the Linux file system with window NT file system. 03
- (d) A disk contains 200 tracks from (0-199). Request queue contains request no. in the following sequence: 98, 183, 37, 122, 14, 124, 65, 67 and current position of the head is 53. Calculate the total no. of tracks movements using FCFS and LOOK algorithm. (assuming head moving towards higher value). 07

OR

Outline the operations that are carried out on a directory. Give advantages and disadvantages of single level, double level and tree structured directory. 07

- Q.5 (a) Define cryptography. 02
- (b) Distinguish between protection and security with example. 02
- (c) Which is more secure Operating System UNIX or DOS? Justify your answer. 03
- (d) Discuss the methods to enforce security in OS. 07

OR

Discuss the OS threats in detail. 07

101-2006
Examination - April/May- 2023
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- Q.1 (a) What are the goals of an Operating System? 02
(b) What is 'Monitor'? Briefly describe its role in Distributed Operating System. 02
(c) What is multiprogramming? How it is different from multiprocessing. 03
(d) Discuss the properties of the following types of operating system: 07
(i) Interactive (ii) Network (iii) Distributed

OR

What is batch operating system? Compare it with multiprogramming OS. Also writes its drawbacks. 07

- Q.2 (a) Explain Binary Semaphores and Counting Semaphores. 02
(b) Why it is important for a scheduler to distinguish between I/O-bound programs from CPU-bound programs? 02
(c) What is critical section problem? How it is solved in OS. 03
(d) Consider the following table:- 07

Process	Arrival Time (ms)	Burst Time (ms)
P1	0	8
P2	1	4
P3	2	9
P4	3	5

Provide the schedule using preemptive shortest job first (SJF).

- (i) Prepare Gantt chart for the above.
(ii) Calculate average turn-around & waiting time for the same.

OR

What are the conditions necessary for the deadlock to occur and methods for handling it? Is it possible that deadlock can occur if one process is running only? Justify. 07

- Q.3 (a) What is swapping? 02
(b) Explain the inverted page table mechanism of memory management. 02



- Q.3 (a) What is swapping? 02
(b) Explain the inverted page table mechanism of memory management. 02
(c) Explain internal and external fragmentation. 03
(d) Find the total no. of page faults and hit ratio using FIFO, Optimal and LRU page replacement algorithm for the given reference string. (Assuming frame size = 3 & 4 and frames are initially empty). 07
Reference String = 2,2,0,1,0,7,0,1,5,3,8,5,7,7,4

OR

A computer system has a 36-bit virtual address space with a page size of 8K, and 4 bytes per page table entry. 07

1. How many pages are in the virtual address space?
2. What is the maximum size of addressable physical memory in this system?
3. If the average process size is 8GB, would you use a one-level, two-level, or three-level page table? Why?

- Q.4 (a) What is an interface? 02
(b) Describe different access methods used for file. 02
(c) How do you choose an optimal technique among the various disk scheduling techniques? Explain. 03
(d) A disk contains 200 tracks from (0-199). Request queue contains request no. in the following sequence: 98, 183, 37, 122, 14, 124, 65, 67 and current position of the head is 53. Calculate the total no. of tracks movements using FCFS and LOOK algorithm. (Assuming head moving towards higher value). 07

OR

Describe the approaches used in free space management? 07

- Q.5 (a) Explain firewall structure in brief. 02
(b) What are the principles of protection? 02
(c) What is user authentication? Give name of methods used to prevent breach of it. 03
(d) How do we use cryptography as a security tool? Explain. 07

OR

What are the security problems in an OS? What role Access Matrix plays in protecting resources in an OS? 07



- Q.9 (a) What is distributed operating system and its types 07
with example.
(b) Explain the concept of remote file access, RPC and
RMI in operating system. 07

OR

- Q.10 (a) What is the major security threats provided by users 07
and providers of computer based system. Explain
what do you understand by a "trusted system".
(b) Write a short notes on the following: 07
(i) Distributed shared memory (DSM)
(ii) Sensor network
(iii) Parallel operating system
(iv) Virtual memory and cache memory

(4)

CS-1152
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Note: Total number of questions are 10. Attempt one question (including all parts) from each unit. Assume missing data, if any, suitably.

UNIT-I

- Q.1 (a) Write a brief note on different operating system 07
structures and distinguish feature of
(i) Real time system
(ii) Multiprocessor system
(b) Discuss the different services provided by an 07
operating system. Explain briefly about various types
of system call provided by an operating system.

OR

- Q.2 (a) Explain the uniprocessor operating system. Give the 07
characteristics differences between a multiprocessor
and uniprocessor operating system.
(b) Describe the Bar machine and batch processing. 07
What meant by spooling and buffering. Explain with
suitable example.

UNIT-II

- Q.3 (a) When a file is saved in computer memory, what type 07
of information gets stored in the directory entry
corresponding to that file? Explain
(b) Discuss various file allocation and access methods. 07
Compare their advantages and disadvantages.

(1)

OR

- Q.4 (a) The requested tracks, in the order received are 55, 58, 39, 18, 90, 160, 150, 38, 184. Apply the following disk scheduling algorithms. Starting track at 100
(i) FCFS (ii) SSTF (iii) SCAN (iv) C-SCAN 07
- (b) Explain the following: 07
(i) Short term scheduler
(ii) Long term scheduler
(iii) Medium term scheduler

UNIT-III

- Q.5 (a) What is process control block (PCB). What is the role of PCB? List the attributes of PCB. 07
- (b) Assume the following process arrive for execution at the time indicated and also mention with the length of the CPU-burst time given in milliseconds. 07

Job	Burst time(ms)	Priority	Arrival time (ms)
P ₁	4	2	0
P ₂	2	2	1
P ₃	3	4	1
P ₄	1	1	2
P ₅	2	3	2

- (i) Give a Gantt chart illustrating the execution of these processes using FCFS, Round Robin (quantum = 1) and priority (preemptive and Non-preemptive)
- (ii) Calculate The average waiting time and average turnaround time for each of the above scheduling algorithm.

(2)

OR

- Q.6 (a) Assume that the average context switching time between processes is S and average amount of time an I/O bound process uses before generating an I/O request is t ($t \gg S$), q is the time quantum. Discuss the affect of each of the following quantum settings:
(i) $q = \text{infinity}$ (ii) $q > t$ (iii) $q = t$ (iv) $q = S$ 07
- (b) Explain in detail how semaphore and semaphore operating can be implemented. Give the solution for readers/ writers problem using semaphore. 07

UNIT-IV

- Q.7 (a) Represent the sentence "we are CSE students" in physical memory if page size is 5 characters and the entries in page table are: 07

0	6
1	5
2	7
3	10

Assume physical memory size to be 55 characters.

- (b) Explain the concept of memory management. Define MFT and MVT with example. 07

OR

- Q.8 (a) Calculate the number of page faults would occur for the following page replacement algorithm with frame size of 4 and 5. Consider the following page reference string:
2, 3, 4, 2, 1, 5, 6, 4, 1, 2, 3, 7, 6, 3, 2, 1
(i) LRU (ii) FIFO (iii) Optimal 07
- (b) Explain the concept of paging. A paging system consists of physical memory 2^{24} bytes, pages of logical address space is 256. Page size of 2^{10} bytes, how many bits are in a logical address? 07

(3)

- Q.9 (a) What is distributed operating system and its types 07
with example.
(b) Explain the concept of remote file access, RPC and RMI in operating system. 07

OR

- Q.10 (a) What is the major security threats provided by users and providers of computer based system. Explain what do you understand by a "trusted system". 07
(b) Write a short notes on the following: 07
(i) Distributed shared memory (DSM)
(ii) Sensor network
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(4)

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(b) Discuss the different services provided by an operating system. Explain briefly about various types of system call provided by an operating system. 07

OR

- Q.2 (a) Explain the uniprocessor operating system. Give the characteristics differences between a multiprocessor and uniprocessor operating system. 07
(b) Describe the Bar machine and batch processing. What meant by spooling and buffering. Explain with suitable example. 07

UNIT-II

- Q.3 (a) When a file is saved in computer memory, what type of information gets stored in the directory entry corresponding to that file? Explain 07
(b) Discuss various file allocation and access methods. Compare their advantages and disadvantages. 07

(1)

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- (b) Write short notes on 03
- (c) Discuss token based and non token based algorithm. 07
- (d) Explain Ricart and Agrawal's multicast based mutual exclusion algorithm. Also state the drawback of algorithm. 07

(1)

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OR

What are key criteria for modeling the distributed deadlock detection and also give suitable example where resource and communication deadlock occur. 07

Discuss the
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- Q.3 (a) Discuss system model in agreement problems. 02
- (b) Write short notes on consensus problem. 02
- (c) What are the issues in distributed file system. 03
- (d) Discuss algorithm for implementation of distributed shared memory. 07

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OR

Explain Andrew file system. 07

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- Q.4 (a) What is checkpoint? 02
- (b) Define commit protocols. 02
- (c) List the difference between forward and backward recovery. 03
- (d) Describe the issue in fault tolerance. 07

OR

Explain voting protocol and dynamic voting protocols. 07

- Q.5 (a) Define concurrency control. 02
- (b) What are locks? 02
- (c) Define timestamps ordering. 03
- (d) What are various implementation techniques for implementation of transaction. 07

OR

Explain two phase locking protocol in detail 07

68/100

- (d) Translate the assignment $x := A[y,z]$ into three address statement.

OR

How syntax directed translation scheme is implemented? Explain with an example.

- Q.4 (a) Draw the typical structure of an activation record.
(b) Under what circumstances does external fragmentation happens.
(c) Explain the concept of type checking and type conversion with an example.
(d) Compare static versus dynamic memory allocation.

OR

Describe in detail about stack allocation of space and heap Management.

- Q.5 (a) Copy propagation leads to dead-code elimination, justify this with example.
(b) What is Peephole optimization?
(c) What is dead code? Explain with an example.
(d) Explain the simple code generator and generate target code sequence for the following statement $d := (a-b) + (a-c) + (a-c)$.

OR

What are basic blocks? Write the algorithm for partitioning into blocks.
